Customer Segmentation & Retention

OBJECTIVE:

The main objective of this project is to analyze customer purchasing behavior, segment customers based on their purchasing patterns, and identify opportunities to improve customer retention and business performance.

PYTHON:

1. Import Libraries and Load Data

In this step, Pandas is used to load and manage the Excel dataset, while Matplotlib is used to create visualizations. The dataset is loaded into a DataFrame, and the first five rows are displayed to understand its structure.

2. Data Cleaning

Data cleaning removes duplicate records and rows with missing CustomerID values. The InvoiceDate column is converted into date format, invalid transactions and cancelled invoices are excluded, and a new Revenue column is calculated by multiplying Quantity by UnitPrice.

3. Customer Analysis Using RFM

The dataset is grouped by CustomerID to calculate Recency, Frequency, and Monetary values. Recency measures the number of days since the last purchase, Frequency counts distinct invoices, and Monetary calculates the total revenue generated by each customer.

4. RFM Scoring

Customers are assigned scores from 1 to 5 for Recency, Frequency, and Monetary value. Recent purchasers receive higher Recency scores, while customers with higher purchase frequency and spending receive higher Frequency and Monetary scores. These scores are combined to create an RFM Score for each customer.

5. Customer Segmentation

Customers are classified into four groups: Champions, Loyal Customers, At Risk, and Regular Customers. The classification is based on their RFM scores and helps identify valuable customers and customers who may need additional engagement.

6. Customer Segment Visualization

A bar chart is created to display the number of customers in each segment. This visualization makes it easier to compare customer groups and understand their distribution.

7. Export the Results

The final customer analysis is exported to a CSV file named customer_segments.csv. The file contains customer IDs, RFM values, RFM scores, and customer segments, making it available for further analysis in SQL Server, Power BI, and Tableau.

Conclusion

The Python module converts raw retail transaction data into meaningful customer insights. It helps analyze purchasing behavior, identify customer segments, and support data-driven decisions for improving customer engagement and retention.

SQL ANALYSIS:

1. Database Creation

Created a SQL Server database named CustomerSegmentationDB to store and analyze the retail transaction data.

2. Data Import

Imported the Online Retail dataset into SQL Server using the SQL Server Import Wizard. The dataset contained approximately 541,909 rows.

3. Data Verification

Checked the imported table names, column names, and total row count to confirm that the data was loaded successfully.

4. Customer Revenue Analysis

Calculated total revenue for each customer using Quantity × UnitPrice and identified the top 10 customers by revenue.

5. Repeat Customer Analysis

Counted the number of unique invoices for each customer and calculated the percentage of customers who made more than one purchase.

6. Purchase Frequency Analysis

Used SQL window functions to calculate the number of days between purchases and estimate the average time between customer purchases.

7. Cohort Retention Analysis

Grouped customers by their first purchase month and calculated the percentage who made additional purchases.

8. RFM Segment Analysis

Imported the customer segmentation results generated in Python into SQL Server. Joined the segmentation table with transaction data to analyze revenue by customer segment.

9. Country-wise Customer Analysis

Grouped customers by country and identified the number of customers and repeat customers in each country.

10. Conclusion

SQL helped analyze customer revenue, repeat purchasing behavior, purchase intervals, cohort retention, and customer segments. These insights can support customer retention strategies and data-driven business decisions.

POWER BI ANALYSIS:

1. Data Connection

Connected Power BI Desktop to the SQL Server database named CustomerSegmentationDB and imported the RetailTransactions and CustomerSegments tables.

2. Data Modeling

Created a relationship between the transaction table and customer segmentation table using the common CustomerID column to enable customer-level analysis.

3. KPI Creation

Created DAX measures to calculate important business metrics, including Total Revenue, Total Customers, Total Orders, and Repeat Customer Rate.

4. Revenue by Customer Segment

Created a bar chart to compare revenue generated by different customer segments, including Champions, Loyal Customers, At Risk, and Regular Customers. Identified Champions as the highest revenue-generating segment.

5. Customer Distribution by Segment

Created a donut chart to display the distribution and percentage of customers across different segments.

6. Monthly Revenue Trend

Created a line chart to analyze monthly revenue from January to December and identify changes in sales performance throughout the year.

7. Repeat Customers by Country

Created a horizontal bar chart to compare repeat customers across different countries and identify the countries with the highest customer retention.

8. Interactive Filters

Added slicers for Month, Segment, and Country to allow users to filter the dashboard and explore specific data.

9. Dashboard Design and Drill-through

Designed an interactive dashboard with KPI cards, four visualizations, and slicers. Created separate detail pages for each chart using drill-through functionality to allow users to explore detailed information.

10. Conclusion

Power BI helped transform the analyzed data into an interactive dashboard. It provides insights into customer segmentation, revenue performance, monthly sales trends, and repeat customer behavior, supporting better customer retention strategies and data-driven business decisions.

TABLEAU ANALYSIS:

1. Data Connection

Connected Tableau to the cleaned raw_data.xlsx and customer_segments.csv files containing retail transactions and customer segmentation data.

2. Data Modeling

Created a relationship between the Online Retail and customer_segments.csv tables using the common CustomerID column.

3. KPI Creation

Created calculated fields and KPI cards for Total Revenue, Total Customers, Total Orders, and Repeat Customer Rate.

4. Revenue by Customer Segment

Created a bar chart to compare revenue generated by different customer segments, including Champions, Loyal Customers, At Risk, and Regular Customers.

5. Customer Distribution by Segment

Created a pie chart to display the distribution of customers across different customer segments.

6. Monthly Revenue Trend

Created a line chart to analyze monthly revenue and identify changes in sales performance over time.

7. Repeat Customers by Country

Created a bar chart to compare the number of repeat customers across different countries.

8. Interactive Filters

Added filters for Segment, Country, and InvoiceDate to allow users to explore specific customer and sales information.

9. Dashboard Design

Created an interactive dashboard titled Customer Segmentation & Retention Analysis Dashboard containing four KPI cards and four visualizations. Applied formatting, sorting, labels, and filters for better presentation.

10. Conclusion

Tableau helped transform the analyzed retail data into an interactive dashboard. The dashboard provides insights into customer segments, revenue performance, monthly sales trends, and repeat customer behavior, supporting better customer retention strategies and data-driven business decisions.

RECOMMENDATIONS:

- focus on champions by providing loyalty rewards and special offers.
- retain loyal customers through personalized promotions.
- re-engage at-risk customers using discounts and targeted campaigns.
- encourage regular customers to purchase more frequently.
- monitor monthly revenue and repeat customers to improve business performance.

CONCLUSION:

the project analyzed customer behavior using rfm analysis and divided customers into different segments. the tableau dashboard provides clear insights into revenue, customer segments, sales trends, and repeat customers. these insights can help the business improve customer retention and make better business decisions.